BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

ASSIGNMENT DOCUMENTATION
SOFTWARE SYSTEM SPECIFICATION & ARCHITECTURAL MODELING

Item	Details
Course	SS ZG562: Software Engineering and Management
Project Title	Smart Construction Site Material & Inventory Management System
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Submitted as a group assignment

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1. Executive Summary, Case Study and Scope

Civil construction projects frequently experience material wastage, theft, stockouts, manual data-entry errors and procurement delays. These issues are intensified across multiple construction sites where connectivity may be intermittent and material records are maintained on paper. The Smart Construction Site Material & Inventory Management System digitizes material requests, purchase orders, delivery verification, inventory updates and reorder decisions in one auditable workflow.

Target Indicator	Target
Material wastage and theft	20% reduction within six months
Stockout-induced work stoppages	35% reduction
Purchase-order approval to dispatch	Less than four hours
Physical-to-digital inventory accuracy	98% match

1.1 In-Scope Functions

- Material requisition and purchase-order management.
- Supplier and purchase-order status tracking.
- QR-code delivery-docket verification.
- Recording delivered quantity, batch number and storage yard.
- Inventory balance updates and audit logging.
- Low-stock detection, reorder recommendations and alerts.
- Offline transaction capture using encrypted SQLite storage.
- Automatic synchronization after network restoration.
- Photo evidence for weighbridge slips and quality documents.
- Role-based access for Site Engineers, Site Managers, Suppliers and Project Directors.

1.2 Out-of-Scope Functions

- Payroll and attendance management.
- Complete accounting and financial-ledger management.
- Construction design and structural analysis.
- Fleet maintenance and vehicle tracking.
- Supplier payment processing.
- Automated physical movement of material inside the site.

1.3 Assumptions

- Each site has a unique site identifier.
- Materials have configured minimum-stock and reorder quantities.
- Approved suppliers provide valid purchase-order information.
- Site Engineers have QR-capable mobile devices.
- Site Managers are authorized to approve purchase orders.

2. Business Goals and Non-Functional Requirements

Goal ID	Business Objective	Target Impact / Metric
BG-01	Material Wastage Reduction	Reduce wastage, spoilage and theft by 20% within six months.
BG-02	Downtime Prevention	Decrease stockout-induced work stoppages by 35%.
BG-03	Procurement Efficiency	Reduce approval-to-dispatch lead time from 48 hours to less than four hours.
BG-04	Inventory Reconciliation	Achieve 98% agreement between physical audits and digital logs.

NFR ID	Quality Attribute	Specification and Verification Criterion
NFR-01	Availability	99.9% uptime during operational hours (06:00-22:00).

NFR ID	Quality Attribute	Specification and Verification Criterion
NFR-02	Performance	Stock-balance queries within 2 seconds under 500 concurrent users.
NFR-03	Offline Synchronization	SQLite stores 1,000 offline transactions and synchronizes within 30 seconds of reconnection.
NFR-04	Security and Integrity	TLS 1.3, AES-256 local storage and RBAC.
NFR-05	Usability	High-contrast UI with touch targets of at least 48 px.

3. Use-Case Model

The use-case model defines the interaction between operational actors and the system boundary.

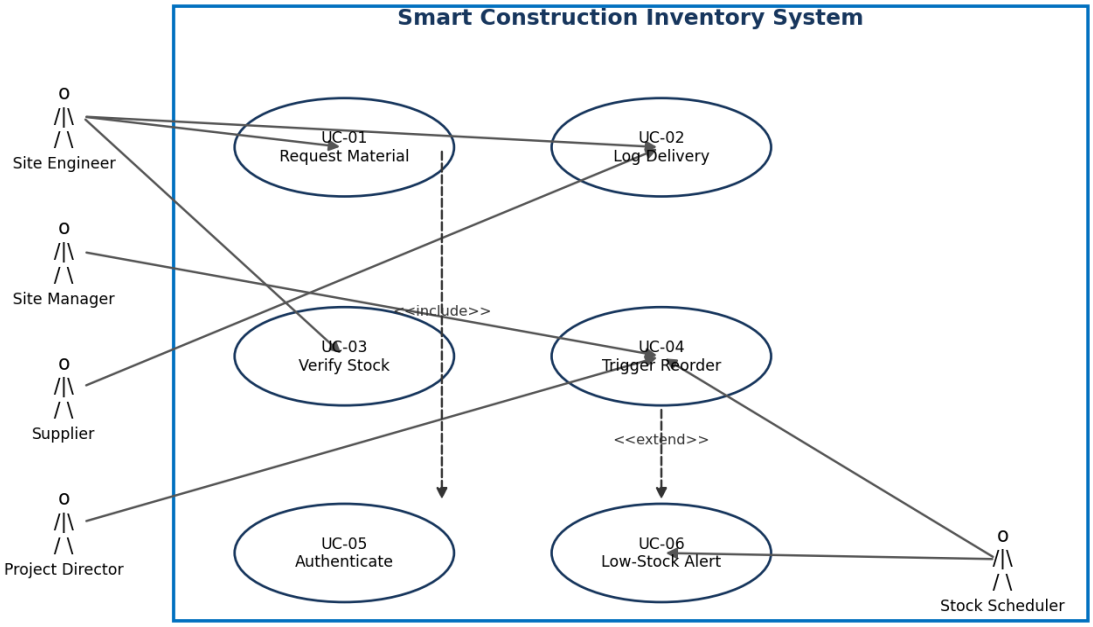


Figure 1. Use-Case Diagram and System Boundary

Actor	Responsibility
Site Engineer	Raises requests, records deliveries and verifies physical shipments.
Site Manager	Reviews stock, approves POs and handles exceptions.
Supplier	Acknowledges POs and dispatches material.
Project Director	Provides second approval for high-value POs.
Stock Scheduler	Checks thresholds and initiates reorder recommendations.

4. Use-Case Prioritization

Priority	Use Case	Rationale	Release
Must Have	UC-05 Authenticate User	Security baseline and RBAC.	Sprint 1
Must Have	UC-02 Log Material Delivery	Core receipt and inventory update.	Sprint 1
Must Have	UC-03 Verify Stock Level	Accurate stock information.	Sprint 1
Must Have	UC-04 Trigger Reorder	Prevents stockouts.	Sprint 1
Should Have	UC-01 Request Material	Automates approval workflow.	Sprint 2
Should Have	UC-06 Low-Stock Alert	Proactive notification.	Sprint 2
Could Have	Supplier Performance Rating	Future supplier analytics.	Sprint 3

5. Significant Use-Case Specifications

5.1 UC-02: Log Material Delivery

Template Field	Specification
Goal	Record and verify delivery and update inventory.
Primary Actor	Site Engineer / Gatekeeper

Template Field	Specification
Scope and Level	Smart Construction Inventory System; user-goal level.
Trigger	Supplier shipment arrives at the site gate.
Preconditions	User authenticated; PO exists in Dispatched state or local cache.
Main Success Scenario	1. Select Receive Delivery. 2. Scan QR docket. 3. Display PO details. 4. Enter quantity, batch and yard. 5. Attach photo. 6. Validate, create receipt, update inventory and mark PO Delivered.
Alternate Flows	Partial delivery changes PO to Partially Delivered. Quality defect creates a rejection slip and blocks stock addition.
Exception Flow	Offline record is saved to encrypted local storage and synchronized after reconnection.
Postconditions	Inventory, DeliveryReceipt and audit log are updated.
Business Rules	Quantity cannot exceed 105% without manager approval. Weighbridge photo is required for aggregate deliveries above five metric tons.
Special Requirements	QR scanning, offline operation, photo evidence, NFR-03 and NFR-04.

5.2 UC-04: Trigger Reorder and Process Purchase Order

Template Field	Specification
Goal	Create and process a PO when stock reaches the safety threshold.
Primary Actor	Stock Scheduler / Site Manager
Scope and Level	Smart Construction Inventory System; system-goal level.
Trigger	Inventory verification detects stock <= minimum threshold.
Preconditions	Safety stock and reorder quantity configured; approved supplier catalogue exists.
Main Success Scenario	1. Detect low stock. 2. Create draft PO. 3. Notify manager. 4. Verify budget and approve. 5. Transmit PO. 6. Supplier acknowledges and dispatches.
Alternate Flows	Manager adjusts quantity; Engineer requests manual reorder.
Exception Flow	Supplier returns Capacity Exceeded; secondary supplier is suggested.
Postconditions	PO is Approved or Dispatched and approval audit is stored.
Business Rules	Orders above $50,000 require Site Manager and Project Director approval. Reorder is suppressed above 90% yard capacity.
Special Requirements	Dual approval, audit trail, supplier fallback and four-hour target.

6. Software Architecture

A three-tier layered client-server architecture separates presentation, business logic and persistence concerns.

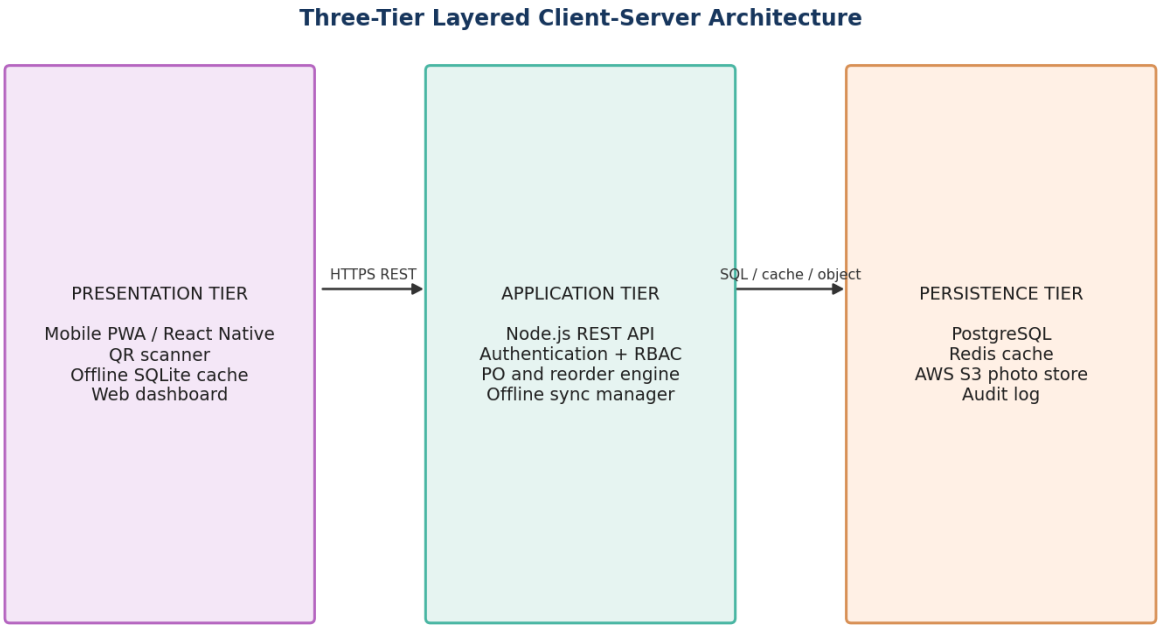


Figure 2. Three-Tier Layered Architecture and Data Flow

Layer	Technology	Responsibilities
Presentation / Client	React Native PWA, SQLite	QR scanning, field UI and local caching.
Application / Middle	Node.js REST API, Express, sync engine	Authentication, RBAC, business rules, reorder logic and synchronization.
Persistence / Backend	PostgreSQL, Redis, AWS S3	Domain entities, stock cache, audit records and photo evidence.

Quality Attribute	Architectural Response
Availability	Redundant API deployment and monitoring.
Performance	Redis cache for frequently accessed stock balances.
Security	TLS 1.3, JWT, RBAC and encrypted local storage.
Offline Support	SQLite queue and reconnect synchronization.
Auditability	User, timestamp, GPS and approval history.

7. Domain Class Modeling

The domain model represents managers, suppliers, materials, purchase orders, delivery receipts and the inventory ledger.

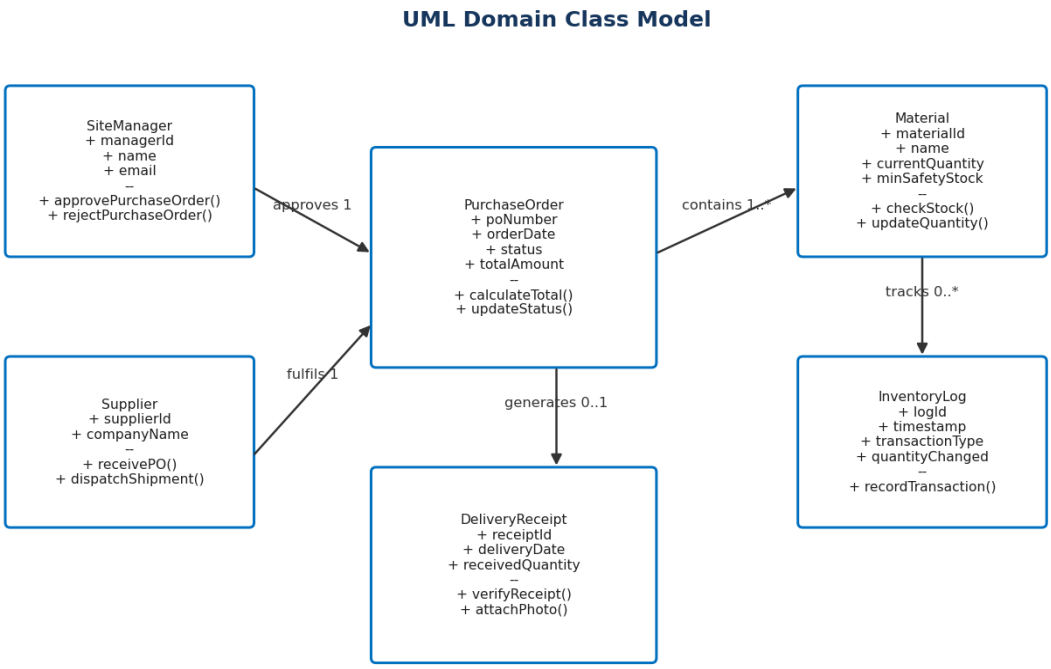


Figure 7. UML Domain Class Model

Class	Key Attributes	Operations
SiteManager	managerId, name, email	approvePurchaseOrder(), rejectPurchaseOrder(), adjustOrderQuantity()
Supplier	supplierId, companyName	receivePurchaseOrder(), acknowledgePurchaseOrder(), dispatchShipment()
PurchaseOrder	poNumber, orderDate, status, totalAmount	calculateTotal(), updateStatus(), requestDualApproval()
Material	materialId, name, currentQuantity, minSafetyStock	checkStock(), updateQuantity(), isBelowSafetyThreshold()
InventoryLog	logId, timestamp, transactionType, quantityChanged	recordTransaction(), getCurrentBalance()
DeliveryReceipt	receiptId, deliveryDate, receivedQuantity	verifyReceipt(), attachPhoto(), markAsRejected()

Relationship	Multiplicity and Meaning
SiteManager approves PurchaseOrder	One manager approves zero or many POs; each PO has an approval record.
Supplier fulfils PurchaseOrder	One supplier fulfils zero or many POs; each PO has one primary supplier.
PurchaseOrder contains Material	A PO contains one or more material lines; a material may appear on many POs.
Material tracks InventoryLog	One material has zero or many inventory transactions.
PurchaseOrder generates DeliveryReceipt	A PO generates zero or many receipts for partial deliveries.

8. Behavioral Modeling

8.1 Activity Diagram

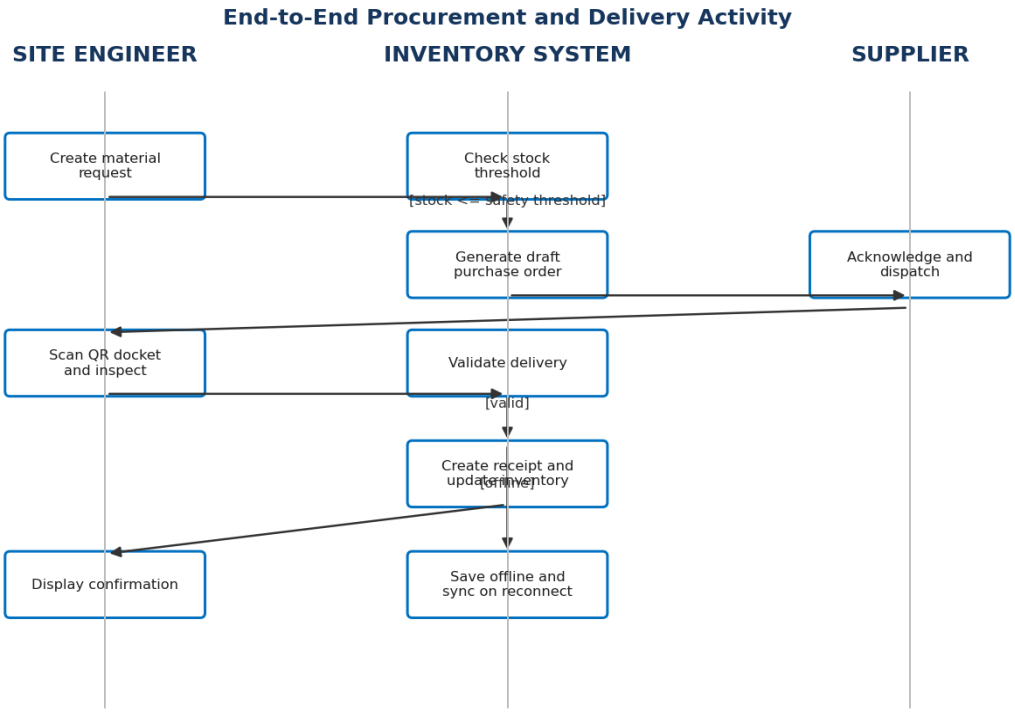


Figure 3. Multi-Swimlane Activity Diagram

8.2 State Transition Diagram

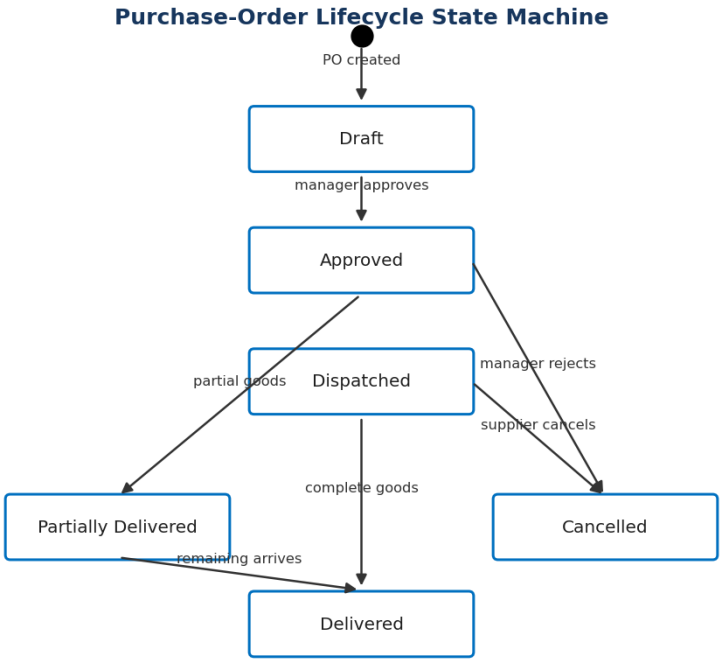


Figure 4. Purchase-Order Lifecycle State Diagram

8.3 System Sequence Diagram

The system-level sequence treats the application as a black box and shows externally visible events for UC-02.

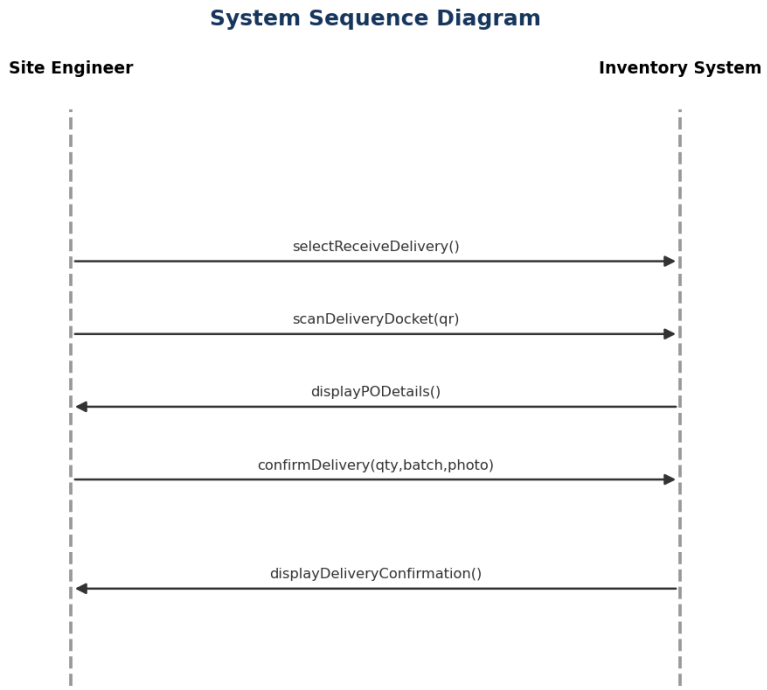


Figure 5. System Sequence Diagram for UC-02

8.4 Class/Object Sequence Diagram

The design-level sequence expands the system event into object interactions and identifies implementable class operations.

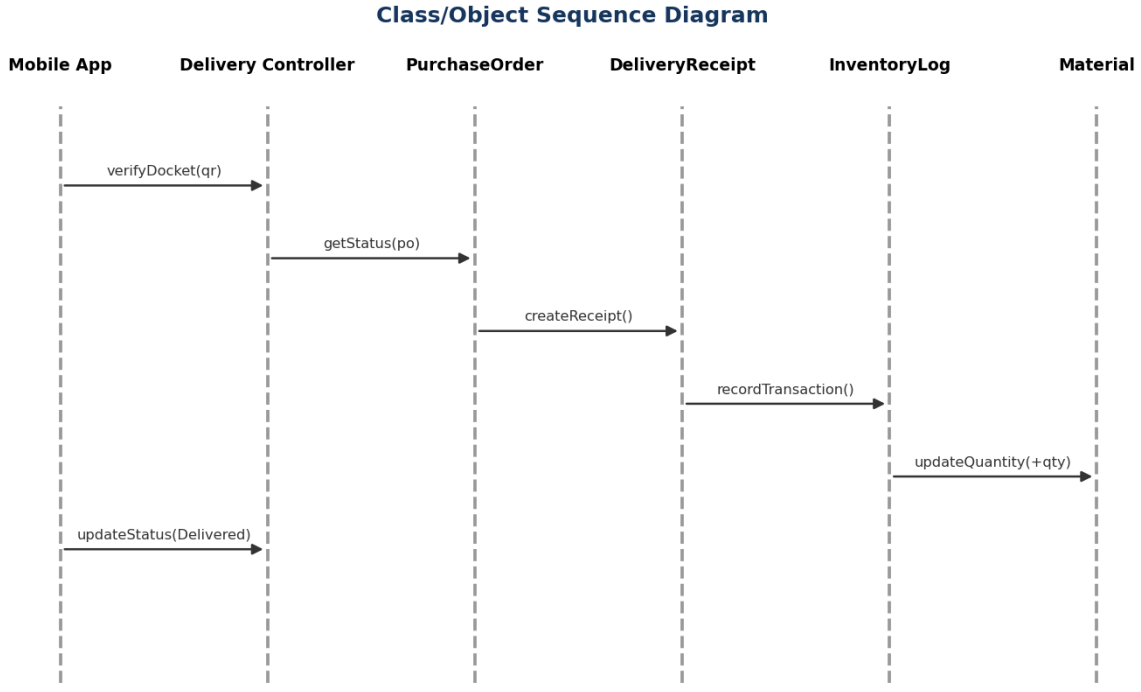


Figure 6. Class/Object Sequence Diagram for UC-02

9. Requirements Traceability and Class-Operation Mapping

Business Goal	Use Case	Domain Class	Sequence Operation	Architecture Layer	Test Case
BG-01 Wastage	UC-02	DeliveryReceipt, InventoryLog	createReceipt(), recordTransaction()	Presentation / Application	TC-DEL-01
BG-02 Downtime	UC-04	Material, PurchaseOrder	isBelowSafetyThreshold(), calculateReorderQuantity()	Application	TC-ORD-02
BG-03 PO Speed	UC-01	SiteManager, PurchaseOrder	createPurchaseOrder(), approvePurchaseOrder()	Application	TC-REQ-03
BG-04 Accuracy	UC-03	Material, InventoryLog	getCurrentBalance(), checkStock()	Persistence	TC-INV-04

9.1 Operation Derivation from the Class Sequence Diagram

Sequence Message	Responsible Class	Purpose
verifyDocket(qrCodeData)	DeliveryController	Validate QR docket and identify the PO.
getStatus(poNumber)	PurchaseOrder	Confirm PO eligibility for receipt.
createReceipt()	DeliveryReceipt	Create a receipt for delivered quantity.
recordTransaction()	InventoryLog	Record inventory movement for auditability.
updateQuantity(+qty)	Material	Increase available quantity after validation.
updateStatus(Delivered)	PurchaseOrder	Move PO to Delivered state.

9.2 Verification Scenarios

Test ID	Scenario	Expected Result
TC-DEL-01	Valid QR docket and complete delivery	Receipt created, stock increased and PO becomes Delivered.
TC-DEL-02	Offline delivery capture	Encrypted local record saved and synchronized within 30 seconds.
TC-DEL-03	Quantity above 105% of PO	Manager approval required before stock addition.
TC-ORD-01	Stock at safety threshold	Draft PO and low-stock alert generated.
TC-ORD-02	Order above $50,000	Dual approval required before supplier transmission.

10. Conclusion

The Smart Construction Site Material & Inventory Management System provides a complete software-engineering blueprint for controlling construction materials from request through delivery and stock reconciliation. The use-case model defines responsibilities, MoSCoW prioritization aligns delivery with business value, and the two detailed Cockburn-style specifications capture functional and non-functional requirements. The three-tier architecture supports offline field operation, while the class, activity, state and sequence models establish a traceable design. The operation mapping demonstrates that class operations are derived directly from class-sequence interactions.

11. References

- Object Management Group, Unified Modeling Language (UML) Specification.

- Alistair Cockburn, Writing Effective Use Cases.

- SS ZG562 Software Engineering and Management assignment instructions.

- Project domain assumptions and requirements defined by the Site-Sathi group.